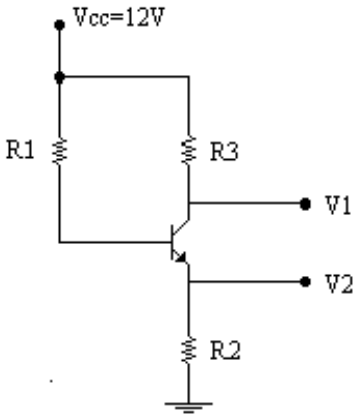
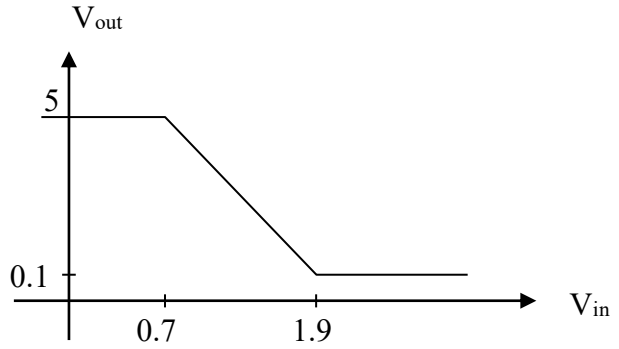
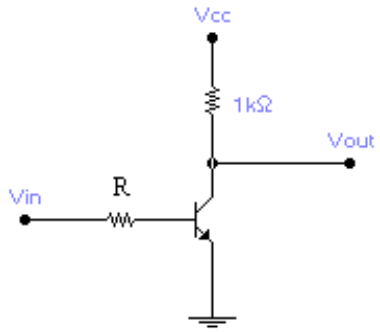
1) For the circuit below, the transistor has β=100, and, R1=100 KΩ, R2=100 Ω, R3=470 Ω. What is the operating region for the transistor? Find transistor currents and V1 and V2 voltages.

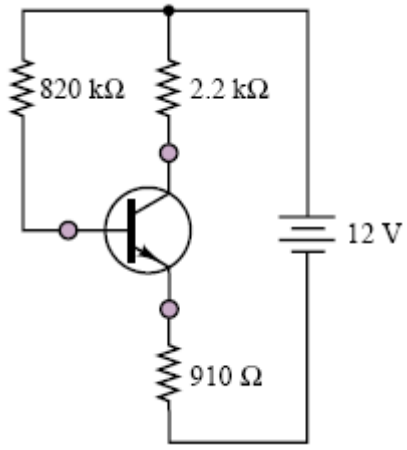


2) We would like to realize the following characteristics with the circuit given. The transistor has β=100. Find $V_{ce(sat)}$, V_{cc}, and R values.

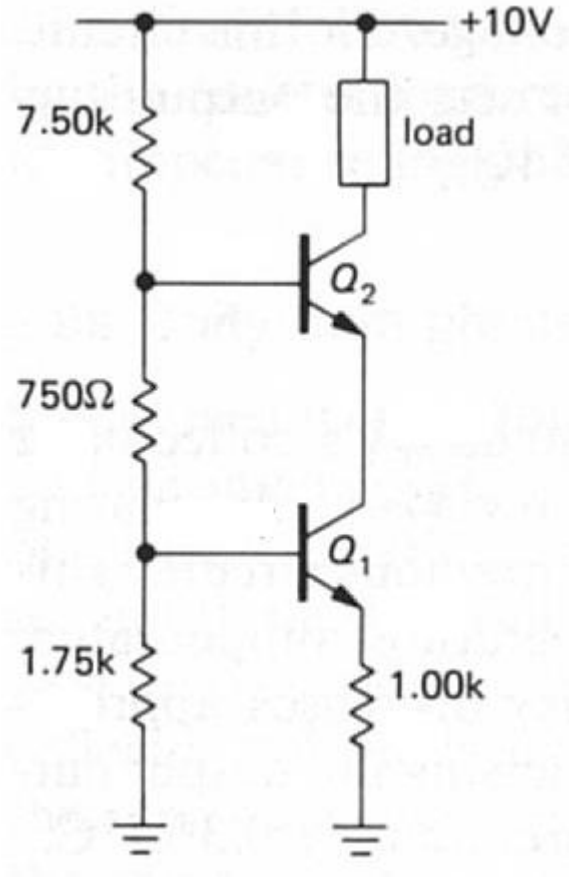




3) Given the circuit below, determine the currents (I_b, I_c, I_e), and, the voltages (V_b, V_c, V_e) of the transistor. Assume the BJT has β = 100. In what region is the transistor?



4) In the circuit below the transistors operate in the active region. Neglect the base currents and find $\mathbf{V_{b1}}$ and $\mathbf{V_{b2}}$. Then find $\mathbf{V_{e1}}$ and $\mathbf{V_{e2}}$. Finally find $\mathbf{I_{e1}}$ and the load current ($\mathbf{I_{c2}}$).
(Note: Since the base currents are neglected, collector currents are equal to emitter currents.)



4) The NMOS transistor shown in the following figure has V_T = 1.5 V, k = 0.4 mA/V^2. If v_G is a pulse with 0 V to 5 V, find the voltage levels of the pulse signal at the drain output.
(That is; find $\boldsymbol{v_D}$ for $\boldsymbol{v_G}$ = 0 and for $\boldsymbol{v_G}$ = 5 V)

